Vadla Pavan Kumar

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Summary

Software Engineer with 2+ years of hands-on experience in Java, Spring Boot, and REST API development for enterprise-scale platforms, paired with Angular/TypeScript frontend engineering. Experienced in PostgreSQL schema design, query optimization, and Redis caching to improve application performance, with a consistent record of unit/integration testing, debugging, and reliable production delivery in Agile/Scrum teams. Comfortable working across the full stack, from secure REST APIs to reusable UI components, and collaborating with cross-functional teams to translate business requirements into technical solutions.

Technical Skills

Languages: Java, TypeScript, SQL, C++
Backend: Spring Boot, Spring Security, Spring Data JPA, REST APIs, Microservices
Frontend: Angular, HTML, CSS
Databases: PostgreSQL, MySQL, Redis (Caching)
APIs & Security: RESTful Web Services, JWT Authentication, Authorization, OpenAPI/Swagger
Testing: Unit Testing, Integration Testing
Tools: Git, Maven, Postman, JIRA, Log4j2
Engineering: OOP, Data Structures, SDLC, Agile/Scrum, CI/CD Basics, Performance Optimization

Professional Experience

Tata Consultancy Services Ahmedabad / Gandhinagar, India
Systems Engineer (Software Engineer) *June 2024 – Present*

- Contributed to ICAI's UDIN and CPE enterprise platforms serving 4,000+ daily active users, delivering scalable backend services and production features for business-critical workflows.
- Engineered modular REST APIs using Java/Spring Boot and TypeScript, improving maintainability and enabling consistent frontend-backend integration across systems.
- Optimized PostgreSQL schemas and SQL queries for high-volume transactional workloads, improving database efficiency and application performance.
- Introduced Redis-based caching to reduce repetitive database operations and improve API response times.
- Developed reusable and responsive Angular components, reducing UI duplication and improving consistency across application screens.
- Strengthened frontend-backend integration through secure, JWT/Spring Security-backed REST data exchange, enabling reliable end-to-end workflows.
- Owned development of an internal Project Management module from design through deployment, expanding internal application capabilities.
- Improved application reliability through unit and integration testing, while collaborating with frontend, QA, and client teams in Agile sprints to deliver production features on schedule.

Projects

Online Grocery Ordering Platform (Backend) | *Java 17, Spring Boot 3, PostgreSQL, Redis, JWT, Spring Security* GitHub

- Built a Spring Boot 3 backend (Java 17) for an e-commerce/grocery ordering platform using Spring Data JPA and PostgreSQL.
- Implemented JWT-based authentication and Spring Security to secure REST endpoints.
- Integrated Redis for caching and documented REST APIs with springdoc-OpenAPI (Swagger UI) to support maintainability.
- Structured the project with Maven, Lombok, and Log4j2 for a maintainable backend architecture.

Admin Dashboard System | *Angular, Spring Boot* 2024

- Developed a dashboard to manage user profiles and transaction records with dynamic, maintainable UI rendering.
- Built reusable Angular UI components to ensure consistent design across multiple pages.
- Integrated REST APIs for real-time data updates between frontend and backend.

Education

Vasavi College of Engineering Hyderabad, India
B.E. (Hons.) in Electronics and Communication Engineering, CGPA: 8.79/10 *2020 – 2024*

Publications

Multi-Symbol Detection of QPSK Signal in Noisy Channel | *IEEE Conference Paper, AMATHE 2024*

- DOI: 10.1109/AMATHE61652.2024.10582079

Honors & Awards

On the Spot (Team) Award | *Tata Consultancy Services (TCS), July 2026*

- Recognized for outstanding contribution, dedication, and positive impact within the organization.